



Listen



Stop

Speed

Normal ▼

FACILITATOR COPY · CONTAINS ANSWER KEYS AND GRADING GUIDANCE · DO NOT PUBLISH TO STUDENTS



Optima Academy Online
AN EDUCATION EXPERIENCE COMPANY

Teacher Manual — Integrated Science

Florida Course 2002400 · Canvas Course 68 · High School

Full-year manual covering all 10 modules and both Finals weeks. Built from the live Canvas course (Weeks 1–14 + Semester 1 Finals) and the locally built, not-yet-uploaded Semester 2 files (Modules 6–10 + Week 36 Final).

Regenerated 2026-10-10: Semester 2 content now follows *plan-semester-2-as-built.md*; Semester 1 content is unchanged.

TABLE OF CONTENTS

1. Facilitator copy banner & how to use this manual
2. How this course works
3. Adapting to a Live, 4-Day-a-Week Classroom
4. Ungraded interactives & the module wrapper question
5. AI & VR in this course (student-facing)
6. Using Ignite AI to support students (teacher-facing)
7. Grading categories & Canvas gradebook setup
8. Rubrics for every teacher-graded item
9. Quizzes, assignments & discussions — how to run & grade
10. Coach, not catch
11. Weekly announcement scripts
12. Teacher workflow & workload map
13. Standards coverage map
14. What makes this course classical & cultivating culture

0. Facilitator copy · how to use this manual

This is a teacher-only document. It contains rubrics, answer guides, grading thresholds, and sensitive-content scripts that students must never see. It is never published to the student-facing Canvas course. Keep it as a standalone PDF/HTML pair alongside the course build, not inside Canvas.

What this manual documents, and how it was built

This manual was written by reading the actual built course, not by assuming what should be there. Two different states exist side by side right now, and the facilitator needs to hold both in mind:

- **Modules 1–5 and the Semester 1 Finals (Weeks 1–15)** exist as real Canvas objects in Course 68 — 15 Canvas Modules, ~100+ Pages, 56+ Assignments, 15 Quizzes — confirmed live via the Canvas API. Every one of these objects is currently **unpublished**.
- **Modules 6–10 and the Semester 2 Final (Weeks 16–36)** exist only as local HTML files in `03_Develop\module-06-evolution-classification` through `module-10-universe` and `semester-2-finals\`. None of this content has been pushed to Canvas or kitted for teachers. A facilitator who logs into Canvas today will see only Modules 1–5 and the Semester 1 Finals; Semester 2 does not yet exist as a Canvas object anywhere. This manual documents Semester 2 fully from the local files so the facilitator has the complete picture before that upload happens, but every fact in Sections 1–13 that concerns Modules 6–10 or Week 36 is a statement about *files on disk*, not about anything a student or facilitator can currently open in Canvas.

Semester 2 as built (regenerated 2026-10-10). Every Semester 2 statement in this manual follows `02_Design\integrated-science-skill\plan-semester-2-as-built.md`, which was written from the built files and supersedes `plan.md` for Weeks 16–36 (`plan.md` remains the source for Semester 1). Built and checked against the folders: 20 instructional weeks, each with 4 readings, 4 lessons, 4 assignments and 1 reading quiz (80 readings, 80 lessons, 80 assignments, 20 quizzes); 5 modules each with a CER, a Writing assignment and a VR Snapshot (15 module-end files); and the Week 36 finals (Review Lesson, Review Activity, Form A Final Exam). Module map: Module 6 Weeks 16–19, Module 7 Weeks 20–23, Module 8 Weeks 24–26, Module 9 Weeks 27–31, Module 10 Weeks

32–35, Finals Week 36. Module-end items are due at the end of Weeks 19, 23, 26, 31 and 35. **Still open for Semester 2:** the files are local only (not in Canvas, not kitted); each reading quiz is worth 10 points (1 per question) with two attempts, highest score kept; the approved Canvas group weights in Section 5 are not yet set in Canvas.

Build-process gap 1 — no course CLAUDE.md. Unlike other Optima courses, this course folder contains no `CLAUDE.md` identity file. The closest equivalents are `02_Design\integrated-science-skill\SKILL.md` (mission and tone) and `02_Design\integrated-science-skill\plan.md` (the actual week-by-week, CER, VR, virtue, and quiz-count specification this manual draws from). Notably, `semester-1-finals\finals-quiz-spec.md` (line 12) states that reading-quiz questions were written to function as standalone finals-bank items "*per CLAUDE.md specification*" — a reference to a file that does not exist anywhere in this course's folder tree. Treat this as leftover boilerplate carried over from a template course, not as a missing deliverable to chase down.

Build-process gap 2 — the syllabus unit map and grading categories (resolved in the regenerated syllabus; not yet live in Canvas). The syllabus originally published in Canvas (`GET /courses/68?include[]=syllabus_body`) described the course as **14 units** (an unbuilt one-week "Bridge Unit: Earth in Cosmic Context" and a split of what the Course Map calls Module 10 into "The Sun" and "The Big Bang") and listed five grading categories, including a 15% "Socratic Seminar and Participation" category for which no discussions were ever built. The syllabus was regenerated on 2026-10-10 to the **10-module map** used throughout this manual, and its old grading categories were replaced by weights that map 1:1 to the six populated Canvas assignment groups (Section 5). Those weights were approved on 2026-10-10, but the Canvas group weights are not yet set. **VERIFY** Confirm that the regenerated syllabus has actually replaced the old one on the Canvas syllabus page before students see it; until then the live syllabus still shows the 14-unit map and the old categories.

Build-process gap 3 — the Semester 2 finals spec (resolved). `semester-2-finals\finals-quiz-spec.md` originally described a 50-question, multiple-choice-only final. It now matches the Course Map: **70 multiple choice (1 point each) + 3 CER (10 points each) = 100 points, 120 minutes, one attempt**, and the exam itself is built as `semester-2-finals\final-exam.md` (Form A). The spec points at the real bank locations under `03_Develop\module-0N-...\week-NN\re`

ading-quiz.md . What remains is building Form A into a Canvas Quiz (Section 1A, Week 36).

Everywhere else in this manual, a claim about the course is grounded in a specific artifact: an API response, a local HTML file, or `plan.md` . Anything not directly verifiable is marked

VERIFY .

1. How this course works

A typical lesson

Every lesson is a standalone, richly formatted HTML page deployed via GitHub Pages and linked into a Canvas Page (this is why lesson pages can carry full custom CSS and small interactive widgets, while anything graded stays Canvas-native — see Section 2). Walking **Week 16, Lesson 1: "What Bones and Stone Tell Us About Life's History"** (Module 6) top to bottom is representative of every lesson in the course:

1. **Hero header** — week/lesson tag, a human-voiced title, and a one-paragraph subtitle framing the lesson's question.
2. **Learning objectives card** — 3–4 objectives in "I will be able to..." form.
3. **History-of-science callout** (blue) — here, Lamarck's 1809 theory: right about the fact of evolution, wrong about the mechanism, and the intellectual courage the era demanded.
4. **Core content** in card blocks with subheadings (here: reading the fossil record, transitional fossils, homologous and vestigial structures) plus an embedded diagram (SVG, not an external image).
5. **Contemporary-research callout** (gold) — hedged, e.g. "Researchers have now sequenced..." on ancient DNA and paleogenomics.
6. **Two ungraded self-check widgets** — a "Pause and Reflect" multiple-choice-with-feedback prompt, and a "Check Your Understanding" true/false set. Both are small inline JavaScript widgets that give the student immediate feedback and submit nothing to Canvas.
7. **VR callout** (green) — "Your VR Session This Lesson," describing what to build/observe in-headset and what evidence to capture, formative only.
8. **Virtue callout** (tan/gold) — ties the lesson's content to the module's virtue (here, Courage).

9. **"The Long View" closing panel** — states the cross-module connection and previews where the module's CER assignment will pick the thread back up.

Every lesson in every module follows this shape; only the content, the callouts present (history/contemporary/AI/VR are included "where appropriate," not mechanically every time), and the graded touchpoint change. The graded touchpoint is never on the lesson page itself — it is the paired Assignment, linked from the lesson but submitted and graded through Canvas.

A typical week

Segment	What the student does	Graded?	Tier
4 Readings (Pages)	Adapted content reading, one per lesson topic	No	Tier 4 (scaffolding)
4 Lessons (Pages, GitHub-deployed)	Core instruction + callouts + in-lesson VR + self-checks	No (self-checks give instant feedback only)	Tier 4 (scaffolding)
4 Assignments (Canvas Assignments)	Short written/analytical response paired to each lesson (10–20 pts each in Semester 1; every Semester 2 assignment is 20 pts)	Yes — teacher/rubric-reviewed	Tier 1–2 (see Section 5)
1 Reading Quiz (Canvas Quiz)	10 standalone multiple-choice questions covering all 4 readings	Yes — auto-graded	Tier 1
VR sessions (in-lesson callouts)	4 formative VR activities, one per lesson, building toward the module VR Snapshot	No	Tier 4

At the end of every module (the module's last instructional week), three additional items land: the **Module CER Assignment**, the **Module Writing Assignment**, and the **Module VR Snapshot**. This is why module-ending weeks show 16 Canvas module items instead of 13 in the live course (confirmed via the Canvas API: Weeks 2, 4, 6, 10, and 14 each carry 16 items; every other normal week carries 13). In the Semester 2 build the module-end weeks are Weeks 19, 23, 26, 31 and 35 (5 modules × 3 module-end items).

Major assessments

Item	Where	Tier	What it asks	Teacher time (per item, 500 students)
Module CER Assignment (×10)	End of each module	3	Claim-Evidence-Reasoning response (350–500 words in Semester 2); 40 pts in Semester	~40 hrs

2, 60 pts in Semester 1 (as built), rubric-scored				
Module Writing Assignment (×10)	End of each module	3	Analytical response, not a summary (350–500 words in Semester 2); 30 pts in Semester 2, 60 pts in Semester 1 (as built), rubric-scored	~40 hrs
Module VR Snapshot (×10)	End of each module	3	Compiled VR snapshot + reflection, 50 pts in Semester 2, 100 pts in Semester 1 (as built), rubric-scored	~35–40 hrs (snapshot review is faster than reading prose)
Weekly Assignments (×56 in Canvas; ×80 more built locally for Semester 2)	Every week, 4/week	1–2	Short paired response per lesson, 10–20 pts (20 in Semester 2), simple rubric	Minutes each; batchable
Reading Quiz (×14 in Canvas; ×20 more built locally for Semester 2, 200 questions)	Every week	1	10 standalone MC questions, auto-graded, 10 points (1 per question), two attempts with the higher score kept	0
Semester 1 Final Exam (built, Quiz ID 1358)	Week 15	3 (partial)	50 MC, auto-graded, drawn from the 140-question S1 bank. No CER items currently exist despite the Course Map calling for 2.	0 (MC only, currently)
Semester 2 Final Exam (Form A built locally as final-exam.md ; not yet in Canvas)	Week 36	1 + 3	70 MC (1 pt each) + 3 teacher-graded CER (10 pts each) = 100 pts; 120-minute limit, 1 attempt, score-only display. CER scoring guides are in the exam file (Section 6). See Section 0, gap 3	~30 hrs for the 3 CER items

1A. Adapting to a Live, 4-Day-a-Week Classroom

This course was designed for fully asynchronous delivery, but the same content works for a facilitator meeting students four days a week. Below is one weekly pattern per module — not per individual week — built directly from the actual weekly shape documented in Section 1 (4 readings, 4 lessons, 4 assignments, 1 reading quiz, 4 in-lesson VR callouts), redistributed across four live class days. Where a module spans multiple weeks, repeat the

pattern each week; the note under each module says where that module's CER, Writing, and VR Snapshot land in the live calendar.

Generic weekly rhythm (the pattern every module below customizes):

Day	In class
Day 1	Readings 1–2 (assigned as advance reading or done silently first 10 min) → Lessons 1–2 delivered/discussed → warm-up discussion built from that week's history-of-science callout → start Assignment 1.
Day 2	Readings 3–4 → Lessons 3–4 → VR session time (rotate headsets or run as a station; the four in-lesson VR callouts can be compressed into one class VR block) → Assignment 2 work time.
Day 3	Assignment 3–4 work time → peer workshopping of any module-level CER/Writing draft in progress → teacher circulates for individual feedback.
Day 4	Reading Quiz (10 Q, timed in class) → wrap-up discussion connecting the week's four lessons → preview next week.

► **Module 1 — Nature of Science (Weeks 1–2)**

► **Module 2 — Atomic Theory (Weeks 3–4)**

► **Module 3 — Macromolecules and Properties of Water (Weeks 5–6)**

► **Module 4 — Cells: Structure, Function, and Energy (Weeks 7–10)**

► **Module 5 — Genetics and Heredity (Weeks 11–14)**

► **Semester 1 Finals (Week 15)**

► **Module 6 — Evolution, Classification, and Taxonomy (Weeks 16–19)**

► **Module 7 — Ecology (Weeks 20–23)**

► **Module 8 — Geology and Earth Systems (Weeks 24–26)**

► **Module 9 — Newton's Laws of Motion and Energy (Weeks 27–31)**

► **Module 10 — Big Bang, Universe, Sun, and Space (Weeks 32–35)**

► **Semester 2 Finals (Week 36)**

2. Ungraded interactives & the module wrapper question

Ungraded interactives cost you nothing

The self-check widgets embedded in every lesson (multiple-choice-with-feedback prompts, true/false checks, click-to-reveal, matching) are formation-by-practice, not busywork to chase. They produce no Canvas submission and require no grading. A student who skips them shows up indirectly — in weaker Assignment or Reading Quiz performance — never as a missing grade in your gradebook. Don't spend time policing them; spend time reading the downstream signal.

This course does not currently have Canvas-Quiz module intro/outro wrappers. The standard Optima design calls for a survey-mode Canvas Quiz at the start and end of each module (essential question, confidence rating, "what's still unclear," virtue reflection) specifically so a facilitator can triage cohort-level data in 15–20 minutes per module. Neither the live Canvas modules (13 or 16 items per week, all accounted for as readings/lessons/assignments/quizzes/CER/writing/VR — no wrapper quiz among them) nor the local Semester 2 files include a wrapper item. The module's opening framing lives instead inside each module's first lesson, and its closing framing lives inside the Review Lesson/Review Activity at Finals weeks only — there is no per-module closing checkpoint. **VERIFY** If the facilitator wants the early-warning triage move described in Section 10, someone will need to build 10 short survey-mode Quizzes (or accept that this course currently has no per-module confidence/unclear-concept signal at all, and rely on Reading Quiz and Assignment grade trends instead).

Where custom code lives, and where it doesn't

The lesson pages are full standalone HTML documents with their own CSS and small JavaScript widgets, hosted outside Canvas and linked in — this is why they can have interactive self-checks. Every graded item (Assignments, Quizzes, the CER/Writing/VR-Snapshot pages) is instead a plain inline-styled HTML fragment matching the syllabus's Canvas-native visual language (navy header, gold eyebrow labels, Georgia-serif body text) with no script tags, built to paste directly into a Canvas Assignment or Quiz description. That split is deliberate: anything a teacher needs to grade in SpeedGrader has to be a real Canvas object, and this course's assignment/CER/writing/VR-snapshot pages already follow that rule correctly.

3. AI & VR in this course (student-facing)

AI

The plan.md callout calendar includes AI-relevant contemporary-research hooks scattered through the year: AI-assisted data analysis and preregistration (Module 1), quantum computing (Module 2), mRNA vaccine technology and CRISPR (Module 3), organoid research (Module 4), CRISPR gene editing and epigenetics (Module 5), ancient-DNA/paleogenomics (Module 6, confirmed directly in the Week 16 Lesson 1 file), AI image classification across the EM spectrum (a plan.md hook for Module 9; the as-built Semester 2 Module 9 callouts are listed in `plan-semester-2-as-built.md` and do not include an AI-specific one, so treat this as not built **VERIFY**). These are hedged, factual callouts about what AI can and cannot do for a given scientific task — not a requirement that students use any particular tool. There is no separate, dedicated "AI literacy" callout type documented in this course's plan the way Optima's Biology course uses one; AI appears here folded into the contemporary-research callout slot. Do not add new student-facing AI requirements beyond what the built lessons already contain.

VR

VR in this course is lighter-touch than some other Optima science courses: each lesson carries one ungraded "Your VR Session This Lesson" callout (4 per week, formative, described by what the student builds or observes and what evidence to capture — see `plan.md` Section 3 for the Semester 1 week-by-week VR map and `plan-semester-2-as-built.md` for Semester 2), and each module carries exactly **one** graded VR Snapshot assignment compiling those weekly sessions into a single artifact (submitted as one or two IFX/image captures plus a short written reflection, uploaded to Canvas). In Semester 2 each snapshot is worth 50 points with module-specific panel criteria (Section 6), and the Module 10

snapshot is one submission after Week 35, built over four weekly panels. There is no separate quarterly VR "unit project" layer documented for this course the way Optima's Biology plan uses one — the module-level Snapshot is this course's only graded VR deliverable. Facilitators review VR Snapshots the same way they'd review any Tier 3 submission: against the rubric (Section 6), using the images/description the student submits rather than entering VR themselves.

4. Using Ignite AI to support students (teacher-facing)

VERIFY Ignite AI's exact feature set varies by Canvas release and by what Optima has enabled on this instance. The functional descriptions below assume a standard Canvas AI assistant; confirm against what is actually turned on for Course 68 before promising a specific button to a facilitator.

With ~500 students and one teacher, Ignite AI is what makes individual attention possible at scale. In every case, **AI drafts, the teacher decides** — every output below is something you read, edit, and personally send or post; none of it is ever auto-sent or used to rank, surveil, or penalize a student.

Drafting individualized feedback

For the Module CER, Writing, and VR Snapshot assignments, ask Ignite AI to draft rubric-anchored feedback from a student's submission and your rubric scores (Section 6). It should name what the student did well before naming the gap, in the coach-not-catch voice (Section 8). You edit for accuracy and voice, then post. This is the single highest-leverage use of Ignite AI in this course, since these are the three Tier 3 items graded on every module.

Summarizing quiz analytics and finding sticking points

Reading Quiz item analysis (which questions the cohort missed most) is the fastest way to know what to reteach. Ask Ignite AI to summarize per-question miss rates across a Reading Quiz and flag any question missed by more than ~30% of respondents — that is your signal for the next weekly announcement's "one thing not to miss." Because this course currently has no discussion forums or module wrappers (Sections 2 and 7), Reading Quiz analytics and Weekly Assignment grade trends are the *only* cohort-level signal available until those gaps are closed.

Surfacing students who need a nudge

Ask Ignite AI to flag students with a missing Weekly Assignment, a Reading Quiz score trending down two weeks in a row, or a missed Module CER/Writing/VR deadline, and draft a warm, specific check-in message for each. You send the message you'd actually send — this only saves you the blank-page problem.

Generating practice variants

For a student who needs another rep on a Tier 1 skill (Punnett squares, $F=ma$ problems, balancing chemical equations), ask Ignite AI to generate 2–3 new practice items in the same style as the module's existing Weekly Assignments or Reading Quiz questions, so the student gets a genuinely new problem rather than the one they already saw.

Communicating with families

Use Ignite AI to turn a grade-trend observation into a clear, kind family update, and to translate any family communication when helpful. This pairs directly with the sensitive-content scripts in Section 13 — draft the advance-notice message, then personalize it.

5. Grading categories & Canvas gradebook setup

What exists today

Course 68 already has 7 assignment groups, confirmed via `GET /courses/68/assignment_groups`. **None are weighted** — every group carries `group_weight: 0.0` and the course itself has `apply_assignment_group_weights: false`.

Canvas Group (id)	# Items	What's in it
Assignments (169)	0	Empty, unused — likely a Canvas default group left over from course creation
Weekly Assignments (255)	56	The 4 per-lesson paired assignments × 14 built weeks
Reading Quizzes (256)	14	One 10-question auto-graded quiz per built week
CER Assessments (257)	5	One Module CER per module, Modules 1–5
Writing Assessments (258)	5	One Module Writing Assignment per module, Modules 1–5
VR Snapshots (259)	5	One Module VR Snapshot per module, Modules 1–5

Semester Finals (260)	2	Includes the Semester 1 Final Exam (Quiz ID 1358, 50 MC, 100 pts)
-----------------------	---	---

These counts describe Semester 1 as it exists in Canvas today. Semester 2 will add its items to the same groups when it is uploaded (Weekly Assignments +80, Reading Quizzes +20, CER Assessments +5, Writing Assessments +5, VR Snapshots +5, Semester Finals +2: the Review Activity and the Final Exam).

Syllabus grading categories now match the Canvas groups 1:1 (approved weights)

The syllabus originally published in Canvas listed five categories (CER and Written Reasoning 30%, Unit Assessments 25%, Laboratory Work and Observation Journals 20%, Socratic Seminar and Participation 15%, Final Synthesis Project 10%) that did not map cleanly onto the Canvas groups. The syllabus was regenerated on 2026-10-10 and those categories were replaced with weights that map one-to-one onto the six populated Canvas groups. Laboratory Work, Socratic Seminar and Participation, and the Final Synthesis Project were removed because no labs, seminars or final project are built. **These weights were approved on 2026-10-10. They are not yet set in Canvas.**

Canvas group (id)	Weight	What it contains
Weekly Assignments (255)	30%	The 4 per-lesson assignments each week
Reading Quizzes (256)	10%	One 10-question auto-graded quiz per week
CER Assessments (257)	15%	Module CER, one per module
Writing Assessments (258)	10%	Module Writing Assignment, one per module
VR Snapshots (259)	15%	Module VR Snapshot, one per module
Semester Finals (260)	20%	Final Exam plus Review Activity
Total	100%	The unused "Assignments (169)" group stays at 0%

Because weights apply per Canvas group, each group's share of the grade does not depend on how many points the items inside it carry. The point scale does differ between semesters (see the gradebook map below and Section 6), which teachers will see as different point totals but which does not break group-based weighting. **VERIFY** Decision item: confirm whether the Semester 1 module-end scale (CER 60, Writing 60, VR Snapshot 100, as built in the files) and the Semester 2 scale (CER 40, Writing 30, VR Snapshot 50) should stay different or be aligned before the course goes live. A teacher who wants a participation or discussion grade has no Canvas group for it; a teacher may still run live discussion, ungraded.

Step-by-step Canvas setup

1. Canvas > Settings > Assignment Groups on Course 68 — once the weights are approved, set the weight field on each of the 6 populated groups per the table above; leave "Assignments (169)" unweighted and unused, or delete it if no items are ever assigned to it.
2. Canvas > Settings > check **"Weight final grade based on assignment groups"** (this flips `apply_assignment_group_weights` to true).
3. Set the grading scheme (letter grade or points) to match school policy — not currently set in the API response reviewed.
4. Once wrapper quizzes exist (Section 2), mark them Survey type in the Quiz settings so they don't post a grade at all.
5. Set a course-wide late/missing policy under Gradebook > Settings if the school wants automatic late deductions; none is currently configured.
6. Use completion-scoring (not points) for any item you want credit-for-attempt only, such as VR self-check widgets if they were ever brought into Canvas (they currently are not — see Section 2).

Gradebook map (representative sample — full list is the 100+ items already in Canvas plus the Semester 2 items awaiting upload: 80 readings, 80 lessons, 80 assignments, 20 reading quizzes, 15 module-end items, and the Week 36 Review Lesson, Review Activity and Final Exam)

Item	Canvas Group	Points	Method
Week N Assignment 1–4	Weekly Assignments	10–20 each in Semester 1; 20 each in Semester 2	Rubric (see Section 6)
Week N Reading Quiz	Reading Quizzes	10 in Semester 1; Semester 2 not set VERIFY	Auto-graded MC
Module N CER Assignment	CER Assessments	Semester 1 (Modules 1–5): 60 as built; Semester 2 (Modules 6–10): 40	Rubric
Module N Writing Assignment	Writing Assessments	Semester 1: 60 as built; Semester 2: 30	Rubric
Module N VR Snapshot	VR Snapshots	Semester 1: 100 as built; Semester 2: 50	Rubric
Semester 1 Final Exam	Semester Finals	100	Auto-graded MC (currently; CER items not yet built)

Semester 2 Review Activity	Semester Finals	30	Teacher-graded (typed in Canvas)
Semester 2 Final Exam (Form A)	Semester Finals	100 (70 MC at 1 pt auto-graded + 3 CER at 10 pts teacher-graded)	Auto + rubric

VERIFY The Semester 1 module-end point values above (60/60/100) are taken from the headers of the built Module 1–5 files in `03_Develop` ; the live Canvas objects were not re-read in this regeneration, so confirm the points in Canvas match.

Use the Canvas **module-progress / requirements view** as your at-a-glance signal for ungraded scaffolding completion (readings/lessons opened) — it's the only visibility you'll have into whether students are actually engaging with the non-graded content, since there is no wrapper quiz to ask them directly.

Semester 2 points summary (as built)

Item	Points	Count
Weekly assignments	20 each (80 per week)	80 (4 per week × 20 weeks)
Module CER	40	5
Module Writing	30	5
Module VR Snapshot	50	5
Reading quizzes	10 questions each; 10 points per quiz (1 per question), two attempts, keep the higher score , decided 2026-10-10	20 (200 questions)
Week 36 Review Activity	30	1
Week 36 Final Exam	100 (70 MC at 1 + 3 CER at 10)	1

Weekly assignments by module: Module 6 = 320, Module 7 = 320, Module 8 = 240, Module 9 = 400, Module 10 = 320, for 1,600 points in all. Module-end items total 600 points (5 × 120). Graded items awaiting upload: 80 assignments, 20 quizzes, 15 module-end items, the Review Activity and the Final Exam (117 in all).

6. Rubrics for every teacher-graded item

Zero rubrics currently exist as native Canvas Rubric objects (GET

/courses/68/rubrics returns an empty array). However, the local CER, Writing, and VR Snapshot HTML files already contain fully worked inline rubric tables. The tables below are Canvas-rubric-ready versions of what's already been written — paste each into a native Canvas Rubric attached to its Assignment so it scores inside SpeedGrader, per the standing reminder in optima-canvas-assignments.

Module CER rubrics

The CER point scale and criteria are **not** identical across all 10 modules. Semester 1 CERs (Modules 1–5) are 60 points as built (for example Modules 1, 2, 3 and 5 are a 45- or 50-point Claim/Evidence/Reasoning part plus a 10- or 15-point second part; Module 4 is a two-part CER, Part A photosynthesis at 30 points). Semester 2 CERs (Modules 6–10) are 40 points. **VERIFY** The scale differs between semesters; confirm whether that is intended (Section 5). Module-level CER titles as built:

Module	CER title	Writing assignment title
1	Does This Study Prove What the Headline Claims?	Science Has Been Wrong Before
2	Why Water Is So Strange	An Atom Is Mostly Empty Space
3	Why Cell Membranes Build Themselves	You Are, Literally, What You Eat
4	Light, Oxygen, and Energy: Two CER Arguments	The Mitochondrion Was Once a Free-Living Bacterium
5	From Mutation to Organism: Tracing a Causal Chain	Mutation, Medicine, and Responsibility
6	The Evidence Portfolio for Common Descent	When Evidence Rearranges Everything
7	Is Ecological Restoration Worth the Cost?	The Living World Is a System
8	Was the Scientific Community Right to Reject Wegener?	Fragile or Resilient? Life and the Geological Record
9	Does the Law of Conservation of Energy Prove That Nothing New Can Happen?	The Responsibility of the Engineer
10	Was the Scientific Community Right to Resist the Big Bang?	Are We Alone, and Does It Matter?

Module 6 CER Rubric (40 points)

Confirmed directly from `module-06-evolution-classification\module-06-cer-assignment.html`.

The prompt asks students to argue that four independent lines of evidence (fossil record, comparative anatomy including embryology, biogeography, molecular sequence comparisons) converge on common ancestry, with at least three specific pieces of evidence drawn from at least two different weeks.

Criterion	Full credit	Partial credit	No credit
Claim (8 pts)	Clearly and directly stated; specific enough to be supported by evidence	Present but vague, or significantly modified from the assigned claim (4–7 pts)	No clear claim, or claim contradicts the evidence
Evidence (16 pts)	At least 3 specific, accurate pieces of evidence from at least 2 different lines (fossil record, anatomy, biogeography, molecular); named specifics (fossils, scientists, values, dates)	2 specific pieces, or 3 pieces where one is vague/inaccurate (8–15 pts)	Fewer than 2 specific pieces, or evidence entirely vague/irrelevant
Reasoning (12 pts)	Explains <i>why</i> each piece of evidence supports the claim; does not merely restate evidence; addresses why convergent evidence matters	Reasoning present for some but not all evidence, or restates rather than explains (6–11 pts)	No reasoning, or reasoning is circular
Precision & Honesty (4 pts)	Scientifically accurate; acknowledges what the evidence does and does not establish	Minor factual errors or slight overstatement (2–3 pts)	Significant factual errors or overstated claims about what evidence settles

Module 7–10 CER Rubric (40 points)

Modules 7, 8, 9 and 10 share one rubric shape (confirmed in the four built CER files): Claim 8, Evidence 14, Reasoning 14, Engagement with complexity 4. Length 350–500 words; typed in the Canvas text box. Prompts as built:

Module	What the prompt asks
7	Evaluate the claim that restoration is so expensive and uncertain that money is better spent protecting undegraded habitat; labelled Claim, at least 2 pieces of Module 7 evidence, and Reasoning
8	Was the decades-long rejection of Wegener's 1912 continental drift hypothesis a failure of scientific reasoning or an appropriate response to the evidence then available? Tie the answer to the virtue of Courtesy

9	Is a student's claim correct, partially correct or incorrect that the First and Second Laws mean nothing genuinely new can arise? Evidence must cover conservation, isolated versus open systems, and a concrete new-structure example; reasoning needs a cross-module link (Module 3, Module 4, or stellar nucleosynthesis)
10	Was the 1927–late 1960s resistance to the Big Bang (favoring Steady State) a failure of scientific reasoning or an appropriate response to the evidence then available?

Module Writing Assignment rubrics

Semester 1 Writing assignments are 60 points as built; Semester 2 Writing assignments are 30 points. **VERIFY** The scale differs between semesters. Semester 2 rubrics as built (the Module 6 file is confirmed at `module-06-evolution-classification\module-06-writing-assignment.html`):

Module	Criteria (points)	Prompt in brief
6	Analysis of the Woese episode 12; Connection to a second example 8; Writing quality and precision 10 (350–500 words)	What the Woese episode and the three-domain system reveal about how science works and why intellectual courage is necessary, connected to one other Module 6 example
7	Ecological accuracy 10; Synthesis across the module 8; Argument and specificity 8; Virtue reflection 4 (400–500 words)	Argue that ecological relationships form a system, using 3+ concepts from different weeks and 2+ named examples, ending with a 2–3 sentence reflection on Service
8	Clear argument 8; Geological evidence 10; Cross-module connection 8; Engagement with complexity 4 (350–500 words)	Given five mass extinctions and about 3.8 billion years of persistence, is life fragile, resilient or more complicated, using the time scale plus Module 6/7 concepts
9	Clear argument 8; Physics accuracy 10; Historical grounding 8; Ethical reasoning 4 (350–500 words)	Choose a Module 9 technology, explain the physics, and argue what responsibility its engineer or scientist bears, citing at least one historical Module 9 example
10	Clear argument 8; Scientific evidence 10; Cross-module connections 8; Engagement with uncertainty 4 (350–500 words)	Argue whether life likely exists elsewhere and how that possibility should change how we think about life on Earth

Module VR Snapshot rubrics (as built)

Every Semester 2 snapshot is 50 points with module-specific criteria; do **not** apply a shared base rubric (the base rubric described in `plan.md` Section 6 does not describe the as-built Semester 2 files). Semester 1 snapshots are 100 points as built; their scene elements below are from `plan.md` and were not re-verified in this regeneration. **VERIFY** Check each Semester 1 snapshot's own file before grading.

Module	Required scene elements
1 Nature of Science	3D CER framework (claim + ≥ 2 evidence tiles + reasoning bridge)
2 Atomic Theory	Period 2 atomic model with correct proton/neutron/electron placement
3 Macromolecules	Phospholipid bilayer (≥ 6 phospholipids) + protein channel, labeled
4 Cells	Eukaryotic cell with 6 organelles + photosynthesis/respiration inset
5 Genetics	DNA + transcription-in-progress + translating ribosome, connected spatially

Module (50 pts)	As-built required elements	Rubric points
6 The Tree of Life in 3D	Three-Domain Framework on a Whiteboard IFX (eukaryote origin marked on the Archaea branch, Woese labeled); Organism Gallery of at least 8 IFX from all three domains with at least 2 from each, each labeled with name/domain, a shared derived character and a Module 6 note; Scientist Hall of at least 4 sticky notes (contribution and approximate date); one convergent evolution example. Two snapshots (overview and close-up) plus a 2–3 sentence reflection; about 40–50 minutes in-headset in the persistent Module 6 Session	Three-Domain Framework 12; Organism Gallery 16; Scientist Hall 10; Convergent Evolution Example 8; Reflection and Completeness 4
7 The Ecology of Florida	One scene of four Whiteboard IFX. 1 Energy and Relationships (Florida food web of 6+ organisms, trophic-cascade arrow for apex predator removal, keystone species and 10% rule notes). 2 Biogeochemical and Population (Florida wetland nitrogen cycle with fertilizer disruption, eutrophication note, logistic S-curve with carrying capacity). 3 Biodiversity and Island Biogeography (equilibrium diagram, Florida panther, Florida Wildlife Corridor, extinction debt). 4 Human Impacts and Restoration (Kissimmee before/after/restored, process principle, python or lionfish invasion mechanism, IUCN Red List Florida species). Upload the IFX file or a screenshot	Whiteboards 1–4 at 12 each; Completeness and evidence of process 2
8 Earth's Dynamic Systems	Four Whiteboard IFX panels in one scene. 1 Interior and Plate Tectonics (Week 24): four-layer cross-section, two boundary types with real examples, a note on the evidence that won acceptance after Wegener. 2 Rock and Soil Cycle (Week 25): three rock types, processes, two energy sources, soil O/A/B/C horizons. 3 Geologic Time Scale and Mass Extinctions (Week 26): four eons, three Phanerozoic eras, largest extinction. 4 Climate History (Week 26): 800,000-year glacial/interglacial timeline, three Milankovitch cycles, CO ₂ –temperature note	Panels 1–3 at 12 each; Panel 4 at 10; Evidence of iterative building 4
9 Forces, Energy,	Four Whiteboard IFX panels in one VR Session. 1 Newton's Laws and Gravity (Weeks 27–28): free-body diagram, action-	Panels 1–3 at 12 each; Panel 4 at 10;

and Waves	reaction pair on two objects, $F = ma$ with numbers and units, and a gravity or Kepler orbit diagram. 2 Work, Energy, and Conservation (Week 29): KE-PE bars at 3+ positions, conservation equation with numbers, a friction example tracing "lost" energy. 3 Thermodynamics and Energy Sources (Week 30): three modes of heat transfer, heat engine with Carnot efficiency, two energy sources compared. 4 Waves and the EM Spectrum (Week 31): transverse and longitudinal waves, EM spectrum with 5+ regions, prism and spectroscopy note. One snapshot of the full scene	Evidence of iterative building 4
10 The Universe from Atom to Cosmos	One submission after Week 35 , built over four weekly panels. 1 Stars and the Sun (Week 32): H-R diagram, stellar-evolution flowchart from nebula, a one-sentence nucleosynthesis note. 2 The Solar System (Week 33): eight planets with one fact each, frost line, a subsurface-ocean moon. 3 Big Bang and Cosmology (Week 34): timeline with 5+ events, 5% / 27% / 68% composition, CMB with Penzias and Wilson credited, Hubble's Law $v = H_0 d$ with the recession graph. 4 Galaxies and Earth's Place (Week 35): Milky Way with the Sun at about 26,000 light-years, a space exploration timeline with 6+ milestones, a habitable-zone diagram. The Week 35 capstone scene sits beside the four panels and is not scored separately	Panels 1–3 at 12 each; Panel 4 at 10; Evidence of iterative building 4

VERIFY The Module 8, 9 and 10 criteria point splits above were checked against the built files for Panels 1–2 and against `plan-semester-2-as-built.md` for the rest; spot-check the last two rows of each rubric table in the Canvas rubric when you build it.

What to look for when grading the Module 6 and Module 7 snapshots

These notes were removed from the student-facing snapshot pages and live in `03_Develop\teacher-resources\vr-snapshot-facilitator-notes.md` (for the unpublished Teacher Resources module when the kit is built).

- Module 6.** Students have been building toward this snapshot across four weeks of VR sessions in Weeks 16–19 (extinction timelines, taxonomy diagrams, the three-domain tree, cladograms) in their persistent Module 6 Session. The final session asks them to compile that work into one coherent display, so it is mostly organizational for students who kept their Sessions up, and students who missed sessions need more building time (allow 40–50 minutes in-headset). Key things to look for: **Are the three-domain branches correctly oriented** (Archaea more closely related to Eukarya than Bacteria is)? **Are the convergent evolution examples genuinely convergent**, not homologous? **Are the scientist contributions accurately described?**
- Module 7.** This is the culminating VR assignment for Module 7. Students should have completed all four Whiteboard IFX during their weekly sessions and now consolidate

and expand them into **one organized scene**. Students who completed the weekly sessions will find it straightforward; students who are catching up may need an extended session, and they should use their weekly work as the foundation rather than starting from scratch. The diagram specifications are intended to be achievable within a 40–50 minute session for students who have done the readings and lessons.

Free-response Reading Quiz items

Every Reading Quiz item sampled (Week 1, Week 16) is standalone multiple choice with a stated correct answer — there are currently no free-response reading-quiz questions in this course requiring a teacher answer-guide. The Semester 2 bank is 200 multiple-choice questions (10 per week) in one shared file format, with the answer key balanced A–D (50 each) and a rationale on every question; five questions were swapped on 2026-10-10 to assess the new standards sections (Week 16 Q3, Week 20 Q4, Week 22 Q9, Week 30 Q8 and Q10). Each quiz is worth 10 points (1 per question), with two attempts and the higher score kept. If a future revision adds any, use a compact 2–3 point analytic guide (full credit = names the key idea correctly; partial = names it vaguely or partially; none = absent/wrong) rather than leaving it un-rubriced.

Weekly Assignment rubrics

These already ship inline with each assignment file (confirmed: Week 16 Assignment 1 uses a 3-part table, Full/Partial/No-credit per part, point values summing to the assignment's total; every Semester 2 assignment totals 20 points). No separate generic rubric is needed — each Weekly Assignment carries its own small, part-specific table; attach each as its own native Canvas Rubric when the assignment is published.

Semester 2 Final Exam and Review Activity scoring

Final Exam (Form A, 100 points). Part 1 is 70 multiple-choice questions at 1 point each (auto-graded). Part 2 is 3 CER items at 10 points each, graded by the teacher with the scoring guides in `semester-2-finals\final-exam.md` ; each item is scored Claim 2 / Evidence 4 / Reasoning 4. Students are directed to write about 150–250 words per item. Teacher notes in the file: award credit for correct reasoning even if the parts are ordered differently, and do not penalize arithmetic slips more than 1 point when the method is correct.

CER item	Task	What full credit looks like
CER 1 — Life science (Modules 6–7)	Natural selection versus a neighbor's "learned resistance" claim, using beetle data from a sprayed and a control field	Claim rejects learned resistance in favor of natural selection on pre-existing heritable variation. Evidence cites at least two data points (about 2% in both fields at the start; 94% only in the sprayed field; control stays near 2%). Reasoning uses the

		conditions for natural selection and notes populations, not individuals, change
CER 2 — Physical science (Module 9)	Whether a roller-coaster cart's measured speed (18 m/s) violates conservation of energy	Claim: not violated. Evidence: $PE = mgh = 100,000 \text{ J}$; $KE = \frac{1}{2}mv^2 = 81,000 \text{ J}$; about 19,000 J unaccounted for as motion (1 point per correct value, up to 4). Reasoning: the difference became thermal energy through friction and air resistance
CER 3 — Earth and space science (Module 8)	Seafloor-age data as evidence for seafloor spreading, with a rate calculation and one other independent line of evidence	Claim: data support spreading. Evidence: age pattern, a correct rate (for example 250 km in 5 million years = 5 cm per year), a consistent rate, and a named second line of evidence. Reasoning: youngest rock at the ridge, steady rate implies a continuous process, linked to Wegener's missing mechanism

Review Activity (30 points, typed in Canvas, teacher feedback through SpeedGrader).

Part 1 Cross-Module Connections, 12 points (choose 3 of 6, 4 points each; each response must name both modules and the linking concept precisely); Part 2 Evidence Evaluation, 10 points (five claims, 2 points each); Part 3 Synthesis Reflection, 8 points (150–250 words on prior understanding, evidence, and what changed). This replaces the plan's year-long concept map.

7. Quizzes, assignments & discussions — how to run & grade

By type — general guidance

Type	How it's scored	Teacher move
Reading Quiz (auto-graded)	10 standalone MC, Canvas auto-scores	Skim the item-analysis report after each week closes; if >30% miss one question, address it in the next announcement (Section 9)
Weekly Assignment (short response)	Rubric-scored, 10–20 pts (20 in Semester 2), 2–4 parts each	Batch-grade by part rather than by student — read all Part A responses across a class set, then all Part B, etc.; faster and more consistent
Module CER (Tier 3)	4-criterion rubric (40 pts in Semester 2, 60 pts in Semester 1; Section 6)	Read for the claim first, then check evidence count/specificity, then reasoning; drop 2–3 brief public exemplar comments per module if the LMS setup allows it

Module Writing (Tier 3)	Semester 2: 30 pts, criteria per module (Section 6); Semester 1: 60 pts	Same as CER; watch specifically for retelling-without-analysis, the most common failure mode per the rubric's own full/partial split
Module VR Snapshot (Tier 3)	Semester 2: 50-pt module-specific rubric; Semester 1: 100 pts (Section 6)	Grade from the submitted images + reflection text; you do not need to enter VR yourself. Check the specific accuracy standard for that module (Section 6 table) before scoring
Semester Final	Semester 1: MC auto-graded, CER portion not yet built. Semester 2 (Form A): 70 MC auto-graded + 3 CER at 10 pts each (Claim 2 / Evidence 4 / Reasoning 4)	For Semester 2, grade the 3 CER items with the scoring guides in <code>final-exam.md</code> (Section 6); the Review Activity (30 pts) is graded in SpeedGrader
Discussions	<i>None exist</i>	None are graded; see note below

No discussions are built or graded. Zero discussion topics exist in Course 68 (`GET /courses/68/discussion_topics` returns an empty array). The original syllabus promised a 15% "Socratic Seminar and Participation" category and named specific seminar prompts; the regenerated syllabus removes that category because no seminars are built or graded, and the grading categories now map 1:1 to the Canvas groups (Section 5). A teacher may still run live discussion (Section 1A already assumes it on Day 1 of each week), but it is ungraded and there is no Canvas group for it.

Per-instance table (representative sample)

Module	Item	Type	Full-credit shape
1	Week 1 Assignment 1–4	Weekly Assignment	Distinguishes observation from inference; correctly identifies IV/DV/controls; explains theory vs. law; evaluates a scientific model
4	Module 4 CER (2-part)	Tier 3 CER	Part A cites ≥ 3 data points and explains light-saturation limits; Part B uses both CO_2 measurements and explains why aerobic respiration yields more ATP despite both pathways producing some CO_2
5	Module 5 Writing	Tier 3 Writing	Takes a defended position on whether mutation should be "corrected" via CRISPR; engages the strongest counterargument; uses specific genetics mechanism content
6	Week 16 Assignment 1	Weekly Assignment	Names a transitional fossil with age/location/features; addresses the design-vs-descent alternative; correctly

			uses "homologous structures"; names one vestigial structure and its ancestral function
6	Module 6 CER	Tier 3 CER	≥3 specific pieces of evidence across ≥2 of the four lines {fossil record, comparative anatomy including embryology, biogeography, molecular}; reasoning explains why convergence across independent lines matters
7	Module 7 CER	Tier 3 CER	Takes a position on whether restoration is worth the cost; at least 2 specific pieces of Module 7 evidence (for example the Kissimmee River case); reasoning engages the complexity (cost, uncertainty, protecting intact habitat)
8	Module 8 CER	Tier 3 CER	Judges whether rejecting Wegener was a failure of reasoning or an appropriate response to the evidence available then; evidence is historically specific; reasoning engages the complexity and ties to Courtesy
9	Module 9 CER	Tier 3 CER	Decides whether the claim that conservation means nothing new can happen is correct, partially correct or incorrect; evidence covers conservation, isolated versus open systems and a concrete new-structure example; reasoning links to Module 3, Module 4 or stellar nucleosynthesis
10	Module 10 CER	Tier 3 CER	Judges whether resistance to the Big Bang (Steady State era to the late 1960s) was a failure of reasoning or appropriate given the evidence then available, with the evidence timeline specific

8. Coach, not catch

Your job is to move each student forward, not to detect and punish. In practice:

- **Lead with what's working.** Every rubric comment names the next step before it names the gap.
- **Allow re-attempts on Weekly Assignments** where the school's policy permits it — they're a learning tool, not a one-shot trap.
- **Use Reading Quiz and grade-trend data to find students who need help early**, then reach out warmly (Section 4) — not to build a case against them. This matters more than usual in this course, since there is no wrapper-quiz confidence signal (Section 2) to catch a struggling student before their grades already show it.

- **On suspected academic dishonesty or AI misuse**, open a conversation and ask the student to talk through their thinking rather than leading with an accusation. Treat it as formation, especially on the analytical Writing and CER assignments where AI-generated prose is easiest to lean on.
- **If a facilitator ever adds public exemplar comments** (once discussions exist, or on shared class review of CER drafts), model good thinking — don't single out weak work for public correction.

9. Weekly announcement scripts

Reusable template — clone for any week/module:

Hi everyone — this week we're in [Module N: Title], focused on [one-sentence topic]. The one thing not to miss: [due assessment or tricky concept]. [One warm, specific encouragement naming something the cohort is doing well or should feel confident tackling]. As always, reach out at [contact] with questions. — [Facilitator name]

Drafted scripts, one per module/finals segment

Segment	Script
Module 1, Week 1	Welcome to Integrated Science. This week we start with the question underneath every other question in this course: what does it mean to know something? Your first Reading Quiz opens Friday and covers all four lessons — don't wait until Thursday night. The CER framework you're learning this week (Claim, Evidence, Reasoning) is the tool you'll use in every module for the rest of the year, so take the time to actually understand it now.
Module 2, Week 3	This week we go inside the atom — from Democritus's 2,000-year-old guess to Rutherford's gold foil experiment actually testing it. [Due date] is your Reading Quiz. If the quantum model feels strange, that's the right reaction — it felt strange to the physicists who discovered it too.
Module 3, mid-module check-in	If your confidence on the water/macromolecules material felt shaky this week, you're not alone — this is genuinely dense content. Before [CER due date], revisit the phospholipid bilayer VR session; seeing the self-assembly happen tends to make the concept click faster than reading about it again.

Module 4, pre-CER script	Your Module 4 CER (photosynthesis + fermentation data) is due [date]. This is a two-part response — budget time for both parts, and use actual numbers from the data tables in your evidence, not just the general trend.
Module 5, Week 13 advance notice	Next week's content includes nondisjunction and the genetic conditions that result from it (Trisomy 21, Turner syndrome, Klinefelter syndrome), taught as straightforward biology. If you'd like to preview the material or talk beforehand, reach out anytime.
Semester 1 Finals, Week 15	This week is review only — no new content. The Review Lesson traces the whole semester's arc from atoms to genetics; take real time with it before the Final on [date]. The Final draws from all 14 Reading Quizzes you've already taken, so nothing on it should be a surprise.
Module 6, Week 16 advance notice (send before Week 16 Day 1; repeat a short reminder before Week 19)	This module covers the evidence for evolution and common descent over the next four weeks, and in Week 19 it includes the current scientific thinking on how life may have originated (the Miller-Urey experiment and the RNA World hypothesis). We'll examine the evidence directly and honestly, the same way we examine evidence in every other module. If you have questions or would like to talk before we start, I'm glad to make time.
Module 7, re-engagement nudge	I've noticed a few of you have been quiet the last week or two — totally fine if life got busy, just checking in. Ecology (Weeks 20–23) is a great module to jump back into: the Florida-specific content (the Everglades, coral reefs, the Kissimmee River) tends to land well even if you've fallen a bit behind. Let me know if you need an extension on anything.
Module 7, Week 22 heads-up (send before Week 22 Day 1)	This week and next, the ecology content turns to how humans affect living systems: biodiversity loss, eutrophication and dead zones, and in places the role of a changing climate. As in every module, we stick to the evidence, name what is well established and what is still uncertain, and ask you to reason from data. Reach out anytime with questions.
Module 8, Week 26 heads-up (send before Week 26 Day 1)	This week we study deep time: how scientists date rocks and fossils, the geologic time scale (Earth is about 4.5 billion years old), and the climate record preserved in ice cores and rock. We'll look at how each conclusion is reasoned from evidence, the same way we have all year. Your Module 8 items (CER, Writing, VR Snapshot) are due at the end of this week.
Module 9, Week 30 advance notice (send before Week 30 Day 1)	This week's content includes heat and thermodynamics, and in Lesson 3 nuclear fission and fusion as energy sources — how they work, their tradeoffs, and where the technology stands today. We'll keep the discussion focused on the physics and the evidence, as we do with every topic in this course.
Module 10, Week 32 opener (send before Week 32 Day 1)	This week we open the last module of the year, Big Bang, Universe, Sun, and Space. We start with the scale of the universe, then stars and the H-R diagram, the Sun, and how stars live and die. Your Module 10 VR Snapshot is one submission after Week 35, built over four weekly panels — start Panel 1 this week.

Module 10, Week 34
advance notice (send
before Week 34 Day
1)

This week we turn to the Big Bang and the history of the universe, including the current scientific evidence for its age (approximately 13.8 billion years). As always, we'll look at the evidence directly — the cosmic microwave background, redshift, elemental abundances — and name honestly what remains open, including dark matter and dark energy. Reach out anytime with questions.

Semester 2 Finals,
Week 36

Last week of the year. The Review Lesson closes the loop this course opened in Week 1 — from "what does it mean to know something" all the way to the edge of the observable universe. Take the Review Activity (30 points) seriously; it's designed to show you how much these ten modules actually connect. The Final Exam has 70 multiple-choice questions drawn from your 20 Semester 2 reading quizzes plus three short CER responses; you have 120 minutes and one attempt, so plan a quiet, uninterrupted block. Proud of the work you've put in this year.

10. Teacher workflow & workload map

A normal week, in order

1. Skim Reading Quiz item analysis from the week that just closed (~10 min)
2. Batch-grade the week's 4 Weekly Assignments across the roster, part-by-part (~2–3 hrs/week spread across the course)
3. If a Module CER/Writing/VR Snapshot is due this week, grade it in the module's dedicated grading window (~8–12 hrs spread across the module cycle, not all in one sitting). The Semester 2 module-end grading weeks are Weeks 19, 23, 26, 31 and 35, and Week 36 adds the finals
4. Post the weekly announcement (Section 9)
5. Answer individual messages, prioritizing any student flagged by grade-trend review (Section 4/8)

Because this course currently has no wrapper-quiz triage step (Section 2) and no discussion-grading step (Section 7), the weekly workflow here is lighter on Tier 2 than the standard Optima model assumes — but also has less early-warning visibility. Budget the time you'd normally spend on wrapper/discussion triage into closer attention to Reading Quiz and Weekly Assignment trends instead.

Heavy modules/lessons

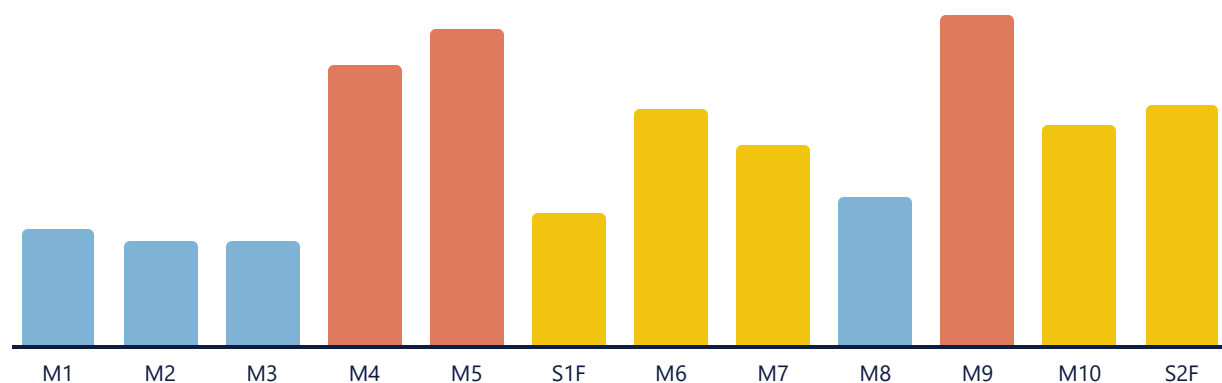
- **Module 4 (Cells, 4 weeks)** and **Module 5 (Genetics, 4 weeks)** — the two longest, densest modules in Semester 1; Module 5 additionally carries sensitive content in Week

13.

- **Module 9 (Newton's Laws and Energy, 5 weeks)** — the longest module in the entire course (400 points of weekly assignments, the most of any module), calculation-heavy throughout. Weeks 29–31 are the heaviest stretch of physics: work, energy, conservation and efficiency in Week 29; thermodynamics and nuclear energy in Week 30; waves and the electromagnetic spectrum in Week 31, with the module-end items due at the end of Week 31.
- **Module 6 (Evolution/Classification, 4 weeks)** and **Module 10 (Universe, 4 weeks)** — moderate-heavy due to sensitive-content framing needs (Section 13) layered onto normal grading load. Module 7 (Ecology, 4 weeks) carries climate-adjacent framing in Weeks 22–23.
- **Module-end grading weeks (Semester 2): Weeks 19, 23, 26, 31 and 35.** Week 26 is the shortest module's end (Module 8, 3 weeks), so its grading follows only three weeks of instruction. Module 10's VR Snapshot is one submission after Week 35 even though students build panels weekly.
- **Both Finals weeks** — Week 15's final has no CER yet; the Week 36 Final Exam has 3 CER items per student (10 points each) to grade with the scoring guides in `final-exam.md`, plus the 30-point Review Activity typed in Canvas, all concentrated in a single week.

Every module carries the same fixed load of 3 Tier-3 items (CER + Writing + VR Snapshot) regardless of length — this is worth flagging on its own: `optima-canvas-assignments` recommends *one* substantive teacher-graded assessment per module to protect the 500:1 arithmetic, and this course runs three every single module. **VERIFY** That is a real structural load a facilitator should plan for, not an error in this manual's math — confirm with course leadership whether that's an accepted tradeoff or something to revisit in a future build.

Workload-fluctuation graphic



Relative facilitator-grading load per module, estimated from module length (weeks) and content density, since every module carries the same fixed count of Tier-3 items (CER + Writing + VR Snapshot). Blue = light · gold = moderate (often sensitive-content prep time,

not just grading volume) · coral = heavy. Plan extra time on the coral modules; the S2 Finals bar reflects the built Form A final with 3 CER items plus the Review Activity (Section 6).

11. Standards coverage map

Standard codes are teacher-facing only. Never let these appear on any student page or on the per-module Canvas Teacher Notes pages beyond this manual's own private copy (Step 5 pages are unpublished, but keep the habit regardless).

By module

Module	Weeks	SC.912 Standards
1 Nature of Science	1–2	N.1.1–1.4, N.1.6, N.1.7, N.2.1, N.3.1, N.3.3–3.5
2 Atomic Theory	3–4	P.8.1–8.5, P.8.7
3 Macromolecules and Properties of Water	5–6	L.18.1, L.18.12, P.8.7
4 Cells: Structure, Function, and Energy	7–10	L.14.1–14.4, L.14.7, L.18.7–18.9
5 Genetics and Heredity	11–14	L.16.1, L.16.14, L.16.16, L.16.17
Semester 1 Finals	15	Cumulative, Modules 1–5
6 Evolution, Classification, and Taxonomy	16–19	L.15.1 (Week 16), L.15.4–15.6, L.15.8 (Weeks 18–19)
7 Ecology	20–23	E.7.1, E.7.3, L.17.2–17.4, L.17.9, L.17.11
8 Geology and Earth Systems	24–26	E.6.1–6.3 (Weeks 24–25)
9 Newton's Laws of Motion and Energy	27–31	P.12.3, P.10.1, P.10.4, P.10.7, P.10.20
10 Big Bang, Universe, Sun, and Space	32–35	E.5.1, E.5.2, E.5.4, E.5.7, E.5.8 (taught in Weeks 31–35; Week 31 is in Module 9)
Semester 2 Finals	36	Cumulative, all 10 modules

Semester 2 benchmark-by-benchmark (as built; checked 2026-10-10)

Source: the CPALMS Integrated Science 1 (#2002400) benchmark list in `01_Analysis\Integrated Science 1 (2002400).pdf`, checked against the built Semester 2 lesson and reading pages by keyword search. This confirms *where* a benchmark is taught; it does not judge depth. The

SC.912 tags inside the Semester 2 files are sparse (18 codes across Modules 6–10), so they cannot serve as a coverage record. Semester 1 rows above were not re-checked here.

Benchmark	Where taught	Status
L.15.1 (evidence for evolution, including comparative embryology)	Week 16 Lesson 1 (reading and lesson)	Covered. A "Comparative Embryology" section was added 2026-10-10, presented as the developmental side of comparative anatomy
L.15.4, L.15.5, L.15.6, L.15.8	Weeks 18–19	Covered
E.5.1, E.5.2, E.5.4, E.5.7, E.5.8	Weeks 31–35	Covered
E.6.1, E.6.2, E.6.3	Weeks 24–25	Covered
E.7.1 (water and carbon cycles)	Week 20, Week 23	Covered
E.7.3 (Earth's systems: atmosphere, hydrosphere, cryosphere, geosphere, biosphere)	Week 20 Lesson 1 (reading and lesson)	Covered. An "Earth's Five Systems and How They Interact" section was added 2026-10-10
L.17.2, L.17.4, L.17.9, L.17.11	Weeks 20–23, 30	Covered
L.17.3 (currents, tides, waves and aquatic abundance)	Week 22 Lesson 2 (reading and lesson)	Covered. A "How Water Movement Shapes Aquatic Life" section with a summary table was added 2026-10-10
P.10.1, P.10.7, P.10.20, P.12.3	Weeks 27–31	Covered
P.10.4 (heat transfer; heat versus temperature and state change)	Week 30 Lesson 1 (reading and lesson)	Covered. A "When Heat Changes Temperature, and When It Changes State" section was added 2026-10-10 (specific heat, $Q = mc\Delta T$, latent heat, heating-curve table)

The four 2026-10-10 sections are new, unreviewed content in the lesson and reading pages. Five reading-quiz questions were swapped the same day so that the new material is assessed (Week 16 Q3 comparative embryology, Week 20 Q4 Earth's systems, Week 22 Q9 upwelling and nutrient supply, Week 30 Q8 specific heat, Week 30 Q10 latent heat); none of those five is in Final Exam Form A, so they are available for a second form. The weekly assignments were not changed.

By standard domain

Domain	Codes carried this year	Modules	Where assessed
--------	-------------------------	---------	----------------

N — Nature of Science	N.1.1–1.4, N.1.6, N.1.7, N.2.1, N.3.1, N.3.3–3.5	Module 1 (foundational; reused as a lens throughout, per the plan.md cross-discipline map)	Weekly Reading Quizzes Weeks 1–2, Module 1 CER/Writing, Sem 1 & Sem 2 Finals (as the CER- reasoning lens across every module)
P — Physical Science	P.8.1–8.5, P.8.7 (Module 2); P.8.7 shared (Module 3); P.12.3, P.10.1, P.10.4, P.10.7, P.10.20 (Module 9)	Modules 2, 3, 9	Weekly Reading Quizzes Weeks 3–6, 27–31; Module 2/3/9 CER, Writing, VR Snapshot; both Finals
L — Life Science	L.18.1, L.18.12 (Module 3); L.14.1–14.4, L.14.7, L.18.7– 18.9 (Module 4); L.16.1, L.16.14, L.16.16–16.17 (Module 5); L.15.1, L.15.4– 15.6, L.15.8 (Module 6); L.17.2–17.4, L.17.9, L.17.11 (Module 7)	Modules 3, 4, 5, 6, 7	Weekly Reading Quizzes Weeks 5–23; every module's CER/Writing/VR Snapshot in this range; both Finals
E — Earth & Space Science	E.7.1, E.7.3 (Module 7); E.6.1–6.3 (Module 8); E.5.1, E.5.2, E.5.4, E.5.7–5.8 (Module 10)	Modules 7, 8, 10	Weekly Reading Quizzes Weeks 20–26, 31–35; Module 7/8/10 CER/Writing/VR Snapshot; both Finals

VERIFY Codes above are transcribed exactly as given in the Course Map inventory this manual was built from. This manual does not have the underlying CPALMS descriptive text for each individual code open for cross-check; confirm exact standard language against CPALMS directly if a facilitator needs it for compliance reporting.

12. What makes this course classical & cultivating culture

What makes it classical

- **The Great Conversation, taught historically.** Nearly every lesson opens with the human story before the mechanism — Lamarck's courageous wrong mechanism (Week 16 Lesson 1), Mendel's 35 years of being ignored, Wegener dying before vindication, Payne-Gaposchkin being told her correct conclusion was "clearly impossible." This is not decoration; it is the course's argument that science is a human inquiry, not a delivered body of facts.
- **CER as applied classical rhetoric.** Claim-Evidence-Reasoning is explicitly Aristotelian argument structure (thesis, proof, logos), taught starting Module 1 and reused as the

single evaluative tool across all 10 modules and both Finals.

- **The virtue thread, tied to a real historical exemplar every module** — not an abstract value statement. Courage is Darwin's 20-year delay and Woese's decade of resistance (Module 6); Perseverance is Rutherford's 20,000 fired alpha particles (Module 2); Wonder opens the course (Module 1, Aristotle/Galileo) and closes it (Module 10, Lemaître/Hubble/Payne-Gaposchkin/Sagan) — a deliberate loop.
- **Wonder before mechanism, and honest acknowledgment of what remains open.**
The syllabus itself states the origin of life and the nature of dark matter/dark energy as genuinely unresolved questions, not settled facts glossed over — consistent with the course's own stated commitment to intellectual honesty as a virtue, not just a slogan.
- **The atoms-to-cosmos spine as the course's actual architecture**, not a marketing description. Week 35 Lesson 4 (students trace how the atoms in their bodies were forged in stars, from atomic theory through to cosmology) and the Semester 2 Review Lesson's "Course Arc" section both make the spine explicit in the built content, not just in planning documents.

The three human threads per module (remapped from units.md)

Module	Virtue	Key exemplars (Modules 1–5 from units.md, remapped to the 10-module structure; Modules 6–10 confirmed against the as-built lessons)
1 Nature of Science	Wonder	Bacon's <i>Novum Organum</i> (1620) argued observation should replace inherited authority; Popper's falsifiability; Kuhn's paradigm shifts; Aristotle's early systematic observation
2 Atomic Theory	Perseverance	Democritus (~400 BCE) imagined atoms 2,000+ years before anyone could test the idea; Dalton, Thomson, Rutherford, Bohr each revised the model as new evidence forced a rethink
3 Macromolecules/Water	Honesty	Wöhler's 1828 urea synthesis quietly ended the belief that organic molecules required a "life force"; Rosalind Franklin's X-ray data and the ethics of how it was used
4 Cells	Self-government	Leeuwenhoek, an untrained draper who ground his own lenses, first saw bacteria; Lynn Margulis's endosymbiotic theory, dismissed then vindicated
5 Genetics	Responsibility	Mendel's eight years of monastery pea-counting, ignored for 35 years; Nettie Stevens's sex-chromosome discovery, credited to a male contemporary

6 Evolution/Classification	Courage	Darwin's 20-year delay before publishing ("Twenty Years of Silence") and Wallace's 1858 letter; Woese's decade defending the rRNA-based three-domain system against establishment resistance
7 Ecology	Service	Humboldt seeing nature as one interconnected system (Weeks 20 and 22); Elton on trophic levels and invasion biology (Weeks 20 and 23); Leopold's land ethic; Carson's <i>Silent Spring</i>
8 Geology	Courtesy	Hutton's "no vestige of a beginning, no prospect of an end" (Weeks 25–26) and Lyell making uniformitarianism the foundation of geology; Wegener, dismissed by geologists and dead before vindication by seafloor-spreading evidence (Week 24; the Module 8 CER)
9 Newton's Laws/Energy	Responsibility	Joule's precise, patient measurement of the mechanical equivalent of heat (Week 29); Julius Robert Mayer, "the idea nobody believed" (Week 29); Lise Meitner, who recognized fission and never received the Nobel Prize (Week 30); and the Module 9 Writing prompt on the responsibility of the engineer. Marie Curie is only mentioned in passing (Week 30 Lesson 3), not a Module 9 exemplar as built
10 Universe	Wonder	Lemaître, a priest and physicist proposing the expanding universe against Einstein's initial resistance (Week 34); Hubble and Leavitt on cosmic distance (Weeks 32 and 34); Payne-Gaposchkin, told by her advisor that her correct conclusion was "clearly impossible" (Week 32); Sagan and the Pale Blue Dot (Week 35)

Cultivating culture asynchronously

- Name good thinking specifically in weekly announcements (Section 9) — not generic praise, but naming what a specific piece of reasoning did well.
- Keep the virtue thread visible every week, not just at module boundaries — the built lessons already do this (every lesson sampled carries its own Virtue callout); echo it in your own announcements.
- Since this course has no discussion forums (Section 7), the CER and Writing feedback you give back is currently the *only* place a real intellectual relationship forms between you and a student's actual thinking. Treat that feedback accordingly — it is carrying more relational weight here than it would in a course with active seminars.

- If you run live discussion (ungraded; no Socratic Seminar is built or graded), use questions with real stakes, such as the Module 8 and Module 10 CER questions about whether the scientific community was right to reject Wegener or to resist the Big Bang.

13. Sensitive-content flags & family/student scripts

No family-notice script currently exists anywhere in the build. The scripts below are new, written for this manual, and designed to be sent in advance of the relevant week — edit the bracketed fields for voice and contact information.

Where	What
Module 6, Weeks 16–19 (full module)	Evolution and common descent as the module's entire subject (Weeks 16–18); human evolution (Week 19 Lesson 3); origins-of-life content (Miller-Urey experiment, RNA World hypothesis) in Week 19 Lesson 4 , not Week 18
Module 5, Week 13	Nondisjunction syndromes (Trisomy 21, Turner syndrome, Klinefelter syndrome) and prenatal genetic testing; sickle cell disease mechanism (Module 5 CER); CRISPR/gene-editing ethics (Module 5 Writing prompt)
Module 7, Weeks 20–23	Climate-adjacent content: climate change appears in Week 22 Lesson 3 (biodiversity loss) and Week 23 Lesson 3 (conservation), eutrophication and dead zones in Week 23 Lesson 1, and sea level in Weeks 20 and 22. Source material is already hedged and non-partisan in framing.
Module 8, Weeks 25–26	Deep time and Earth's roughly 4.5-billion-year age (Week 26 Lesson 1); the geologic time scale and the Anthropocene, global warming and ocean acidification (Week 26 Lesson 3); the climate record and CO2 beyond the ice cores (Week 26 Lesson 4)
Module 9, Week 30	Nuclear fission and fusion as energy sources (Week 30 Lesson 3); energy sustainability and climate (Week 30 Lesson 4)
Module 10, Week 34	Big Bang cosmology and the 13.8-billion-year age of the universe (Week 34; the age first appears in Week 32 Lesson 1); some families flag this alongside evolution content. Week 32 opens the module with the scale of the universe, stars and the Sun

Advance-notice scripts

Module 6 (Evolution/Origins) — send before Week 16 (origin-of-life content arrives in Week 19)

Dear families, next week [student]'s class begins Module 6, which covers the scientific evidence for evolution and common descent, along with, in Week 19, current

scientific thinking on how life may have originated (the Miller-Urey experiment and the RNA World hypothesis). We approach this material the way we approach every topic in this course: examining the evidence directly, naming what it does and does not establish, and treating open questions honestly as open. This is core content for the Florida science standards this course covers. If you'd like to talk beforehand or have any questions about how the material is presented, please reach out to me at [contact] — I'm glad to talk it through.

Module 5 (Genetic testing / CRISPR ethics) — send before Week 13

Dear families, next week's genetics content covers nondisjunction — what happens when chromosomes don't separate correctly during cell division — and the genetic conditions that result, including Trisomy 21 (Down syndrome), Turner syndrome, and Klinefelter syndrome, along with prenatal genetic testing. Later in the module, a writing assignment asks students to think through the ethics of using gene-editing technology like CRISPR to correct inherited mutations. We present the biology plainly and invite students to reason carefully about the ethical questions rather than handing them a settled answer. Please reach out to me at [contact] with any questions.

Module 10 (Big Bang cosmology) — send before Week 34 (Module 10 itself opens in Week 32)

Dear families, next week [student]'s class reaches the Big Bang model in Module 10 (the module began in Week 32 with the scale of the universe, stars and the Sun), covering the scientific evidence for the universe's age of approximately 13.8 billion years — the cosmic microwave background radiation, the redshift of distant galaxies, and the observed abundance of light elements. As with every topic this year, we look at the evidence directly and name honestly what current science does and does not yet explain (including open questions like the nature of dark matter and dark energy). If you have questions about how this is presented in class, I'm happy to talk — reach me at [contact].

Module 7 and Module 8 (climate and deep time) — send before Week 22 and before Week 26

Dear families, over the next few weeks [student]'s science class covers how humans affect ecosystems (biodiversity loss, eutrophication, and in places a changing climate) and, in Week 26, deep time: the roughly 4.5-billion-year age of the Earth and the climate record preserved in ice and rock. We present the evidence plainly, say what is well established and what is still uncertain, and ask students to reason from data. If you have questions about how this is presented, please reach out to me at [contact].